LIGHTYEAR

# MS86 iDempiere boundary findings and operational journeys

Evidence recorded 26 September 2026, following MS85.

**Status: native boundary differences measured; bounded operational equivalence passed under an explicit caller contract. Full application and platform qualification remain unclaimed.**

## Customer benefit

MS85 established one simple customer-to-payment scenario. MS86 deliberately tests values and workflows where a database change can alter business behavior. It produces two concrete findings that the simple journey missed, alongside native evidence for the business cases that do work.

The customer claim is therefore specific: fractional pricing, fixture tax, discount, Unicode and empty optional text have measured application outcomes; fractional shipment time exposes a real Oracle precision loss. A native locking API combination also fails on Oracle. Neither finding is hidden by a passing test-process exit or an alternative caller pattern.

## Boundary journey

The same pinned iDempiere Java application runs on Oracle 26ai Free and PostgreSQL 15.19 using local Docker. A new customer named Café 東京 Łódź orders three units priced at 19.995 before a 10% discount. Actual price 17.9955 produces net 53.99, fixture tax 4.05 at 7.5%, and an invoice/payment of 58.04 after explicit decimal rounding. Stock falls from ten to seven.

Both engines preserve the customer name and turn the empty description into SQL null through the application save path. Independent database readback verifies the amounts, allocation, stock and accounting.

The input shipment timestamp is `2026-09-26T12:34:56.123456`. PostgreSQL preserves it; Oracle's `M_InOut.ShipDate` DATE column stores `2026-09-26T12:34:56`. Oracle passes 14 of 15 explicit checks; PostgreSQL passes all 15. The 913-table comparison retains this field as unresolved. Consequently, `bounded_boundary_equivalence` remains **false**.

This is an application observation, not merely a standalone SQL expression probe. A reviewed timestamp storage/input contract and another native run are needed to close it. Dropping precision from the comparison would not close it.

## Operational journeys

A separate fresh database pair prevents the known timestamp difference from entering these starting states. The harness exercises:

- Two partial shipments: one unit, then the remaining two, followed by an aggregate invoice and payment.
- A one-unit credit memo of 19.35 and a workflow reversal of -19.35, with reciprocal document links and accounting cancellation by account.
- An injected exception before commit, confirmed rollback, a successful retry and an explicit logical-key lookup that prevents duplicate creation.
- Two overlapping, explicitly locked application updates: 100.00 to 100.01 to 100.02, with the second transaction waiting for the first lock.

The first trial used `Query.setForUpdate(true).firstOnly()`. PostgreSQL completed it, but Oracle rejected the generated `FOR UPDATE FETCH FIRST 2 ROWS ONLY` with `ORA-03049`. The failed run, issue rows, native states, harness and error excerpt remain in the evidence archive.

The later harness uses `Query.setForUpdate(true).list()` against a unique primary key and asserts exactly one returned row. Its result applies only to that caller contract. It neither fixes nor qualifies the failed `firstOnly()` combination. The operations comparison records the final verdict separately from the boundary failure.

The final operations run has zero new application issues and zero unresolved differences across 913 common tables. It retains 509 new validated raw field differences. The fractional run retains 311 validated new differences and its one unresolved timestamp difference. Each carries the 5,540 unchanged MS84 differences separately; those are not counted as new coverage.

## Datatype mappings

The new catalog-based inventory covers all 18,852 common base-table columns:

| Mapping or declaration class | Columns |
|---|---|
| NUMBER / numeric | 9,393 |
| Matching declared numeric precision and scale | 8,370 |
| Unbounded or partially declared numeric | 1,023 |
| DATE / timestamp without time zone | 2,259 |
| Text / UUID | 914 |
| CLOB / JSON or JSONB | 3 |
| BLOB / large-object reference | 3 |

The full inventory also includes character, text, binary and timezone-aware timestamp mappings. Counts above overlap by design. Native declarations and one-sided columns remain available. Matching declarations are not behavioral proof: rounding, numeric range, invalid input, multibyte capacity, collation, JSON semantics and large-object ownership remain explicit review obligations.

## Evidence, validation and limits

The complete report links the receipt, datatype review, both Java harnesses and the native evidence archive. Offline tests reconstruct sealed states, validate raw rows and hashes, recompute the outcomes, and retain both failures. CI includes these checks on Windows and Linux without live database credentials.

Local validation ran 212 affected calibration, evidence and documentation tests: 210 passed and two skipped. Milestone manifest verification and whitespace checks passed. All six experiment containers were stopped with data retained.

Six native application executions are retained: two fractional-boundary runs, two runs of the initial operational trial, and two runs of the collection-query alternative. All experiments use the recorded application source commit `731515dcdd5278b843db33b9d3109d155b881951`. Schema repairs preserve the MS84 row multisets before each new experiment.

Controlled two-transaction locking is not load or broad isolation qualification. Pre-commit exception recovery is not crash, ambiguous-commit or failover recovery. Caller-keyed retry is not proof of global application idempotency. One credit reversal is not every reversal type. UUID/JSON boundary

journeys, maximum numeric precision, Unicode length/collation and broader datatype enforcement remain future work. No GCP resources were started; cleanup retains the local data.

## Next work

Resolve the shipment timestamp contract and the Oracle locking-query defect, then rerun their exact failing cases. Extend the established harness to partial invoicing, further reversal types, concurrent business-document processing, deadlocks and failures around commit. Each new claim requires its own declared inputs, independent expected outcomes and retained native observations.